

Problem

A series RLC circuit has the following parameters: $R = 1\text{ k}\Omega$, $L = 1\text{ H}$, and $C = 10\text{ nF}$. What type of damping does this circuit exhibit?

Step-by-step solution

Step 1 of 2

Refer to Figure 8.8 in the textbook.

Calculate the neper frequency, α .

$$\alpha = \frac{R}{2L}$$

Substitute $1\text{ k}\Omega$ for R and 1 H for L in the equation.

$$\begin{aligned}\alpha &= \frac{1 \times 10^3}{(2)(1)} \\ &= 500\end{aligned}$$

Step 2 of 2

Calculate the resonant frequency, ω_0 .

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute 1 H for L and 10 nF for C in the equation.

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{(1)(10 \times 10^{-9})}} \\ &= 10000\text{ rads/s}\end{aligned}$$

Hence, $\alpha < \omega_0$.

Thus, the type of damping is **UNDER DAMPED**.